

THOOGUDEEPA DONNE BIRYANI MANE • ENTERPRISE SUITE

COMPLETE CUSTOMER ENTERPRISE SYSTEM MANUAL & CROSS-ROLE ARCHITECTURE

Comprehensive Operational Flows, Button-by-Button Functionality, Multi-Guest Synchronization, and Cross-Portal Workflows

A definitive, end-to-end technical and operational specification document covering the diner smartphone experience, table QR code session binding, multi-chair ordering, shared billing, real-time KDS integration, Floor Waiter communications, and managerial POS sync.

Repository: github.com/suhassuhas4120/thoogudeepa-enterprise-suite	Active Branch: feature/customer-app
Architecture: Next.js 15, TypeScript, Tailwind CSS, Zustand, TanStack Query	Real-Time Sync: Supabase Realtime & BroadcastChannel('thoogudeepa_bridge_sync')
Document ID: THOOGUDEEPA-CUST-2026-v2.4	Release Date: October 2026
System Scope: Screens 1 to 9 + Waiter, Kitchen, and POS Bridges	Engineering Division: Nelja Restaurant Operating Systems

EXECUTIVE SYSTEM ARCHITECTURE & REAL-TIME HIGHWAY

The Thoogudeepa Customer QR System empowers seated diners to explore the authentic culinary heritage of Donne Biryani Mane, order with fine-grained tossing and spice customizations, monitor kitchen preparation milestones in real time, settle bills through flexible digital and cash tender options, and summon staff on demand. All actions operate on an ultra-low-latency real-time highway connecting mobile diners, Floor Waiters, Kitchen Display Systems (KDS), and Manager Consoles.

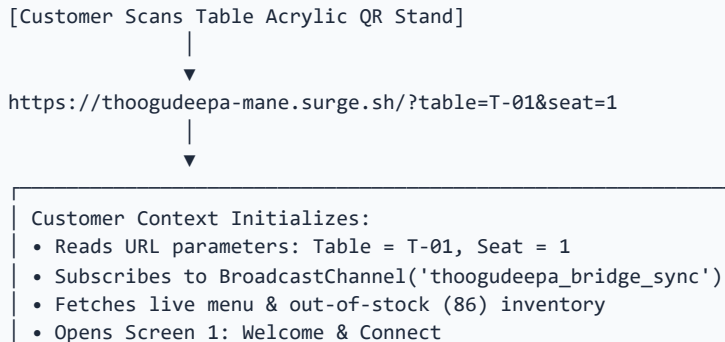
CROSS-ROLE REAL-TIME HIGHWAY BroadcastChannel('thoogudeepa_bridge_sync') & Supabase			
1. CUSTOMER PORTAL (Screens 1 to 9)	2. FLOOR WAITER TABLET (Screens 1 to 6 & Tab)	3. KITCHEN DISPLAY KDS (4 Station Bump Bars)	4. MANAGER CONSOLE (POS & Live Odometer)

Synchronized State Bus: Every client instance listens on BroadcastChannel('thoogudeepa_bridge_sync') and Supabase Realtime websocket subscriptions. Changes made on a customer's phone (such as firing an order, selecting a cash note, or requesting water) dispatch instantly to the Captain's tablet and Kitchen KDS bump bars without polling.

SECTION 1: Self-Customer Connections & Customer-to-Customer Flow (Screens 1 to 9)

1.1 Table QR Scan Onboarding & Multi-Guest Architecture

When guests arrive at Table T-01 and scan the acrylic table stand QR code with their mobile phone cameras, the customer portal initiates an automated onboarding sequence:



Multi-Guest / Customer-to-Customer Table Collaboration:

- **Seat Independence:** Guest A scans the QR code, lands on Screen 1, and selects **Chair C-01**. Guest B scans the same QR stand, lands on Screen 1, and selects **Chair C-02**. Both guests have distinct seat coordinates in their browser state.
- **Order Collaboration & Pooled Bill:** Both guests share the unified Table T-01 order ledger. When Guest A adds Biryani and Guest B adds Chicken Ghee Roast, both items append to the shared table order without collision.
- **Bill Splitting Calculation (Screen 6):** When proceeding to checkout, either guest can toggle the **Split Bill** feature. Selecting 2 Diners, 3 Diners, 4 Diners, or 5 Diners calculates the exact equal share:
$$\text{Per Person Amount} = \text{Math.ceil}(\text{Grand Total} / \text{Number of Diners})$$

Each diner can verify their individual payable share while viewing the consolidated tax breakdown.

SCREEN 1: Welcome & Table Connect (Screen1Welcome.tsx)

Purpose: Authenticates the diner at the table, establishes physical chair assignment, displays restaurant heritage branding, and unlocks the menu via Wi-Fi or cellular network.

Containers & Elements:

- **Container 1: Restaurant Brand Header Card:** Contains Donne Biryani Pot emblem with golden Royal Crown badge, dynamic time-aware greeting badge (Good Morning before 12 PM, Good Afternoon 12 PM–5 PM, Good Evening after 5 PM), establishment title THOOGUDEEPA DONNE BIRIYANI MANE, and heritage tagline Authentic Military Donne Biryani.
- **Container 2: Scanned Table Information Card:** Highlights Table Number (T-01), Chair Tag (C-01), and Live QR Session status badge.
- **Container 3: Interactive Chair Selection Matrix Card:** Houses the 4 dining chair buttons corresponding to physical seats at the table.
- **Container 4: Network Authentication & Gateway Card:** Houses the Wi-Fi and Cellular connect CTA buttons.

- **Container 5: Footer Accreditation Badge:** Displays Powered by Nelja Restaurant Operating System.

Clickable Buttons & Interactive Controls:

1. Chair Selection Buttons (C-01, C-02, C-03, C-04)	
LOCATION: Container 3 (Chair Selection Matrix)	ACTION: Selects active dining seat; updates URL query param ? seat=N via replaceState
STATE CHANGED: seatNumber = N	DESTINATION: Stays on Screen 1 (highlights selected chair)
2. "Connect via 5G Wi-Fi" Button (Hero Orange Button)	
LOCATION: Container 4 (Network Gateway Card)	ACTION: Authenticates diner session with restaurant local network and unlocks digital menu
STATE CHANGED: currentScreen = 2, previousScreen = 1	DESTINATION: Navigates to SCREEN 2 (AUTHENTIC MENU)
3. "Continue via Mobile Data" Button	
LOCATION: Container 4 (Network Gateway Card)	ACTION: Unlocks digital menu using diner's cellular data connection
STATE CHANGED: currentScreen = 2, previousScreen = 1	DESTINATION: Navigates to SCREEN 2 (AUTHENTIC MENU)

SCREEN 2: Authentic Menu & Discovery (Screen2Menu.tsx)

Purpose: Primary culinary catalogue. Provides live text search, dynamic category filtering, instant cart counters, 86-stockout locks, and quick-customization drawer triggers.

Containers & Elements:

- Container 1: Top Navigation Header (WireHeader):** Displays Table Tag (T-01 • C-01), Call Waiter Bell button, and quick Cart icon with red badge counter.
- Container 2: Live Search & Filter Bar:** Search input box with search icon and clear (X) button.
- Container 3: Dynamic Category Tabs Scroller:** Horizontally scrollable pill tabs derived dynamically from live menu data.
- Container 4: Menu Items Grid:** Vertical list of dish cards with food photos, Veg/Non-Veg FSSAI badges, preparation modes, prices, and quantity controls.
- Container 5: Quick Customization Slide-Up Sheet (ItemDrawer):** Bottom sheet modal for in-menu spice and toss selection.
- Container 6: Floating Bottom Cart Bar (Sticky Dock):** Appears whenever `cart.length > 0`, displaying total item count and running bill total.

Clickable Buttons & Interactive Controls:

1. Header Call Waiter Bell Button	
LOCATION: Container 1 (WireHeader Top Right)	ACTION: Opens immediate assistance hub
STATE CHANGED: <code>currentScreen = 9, previousScreen = 2</code>	DESTINATION: Navigates to SCREEN 9 (CALL WAITER)
2. Header Cart Icon Button with Red Counter Badge	
LOCATION: Container 1 (WireHeader Top Right)	ACTION: Opens cart review page
STATE CHANGED: <code>currentScreen = 4, previousScreen = 2</code>	DESTINATION: Navigates to SCREEN 4 (CART REVIEW)
3. Search Clear Button (X icon)	
LOCATION: Container 2 (Search Bar)	ACTION: Resets search input query to empty string
STATE CHANGED: <code>searchTerm = ""</code>	DESTINATION: Stays on Screen 2 (restores full dish catalog)
4. Category Filter Pills (All, Rice & Bowls, Starters, Desserts, Chef Special)	
LOCATION: Container 3 (Category Tabs)	ACTION: Filters visible dishes to the clicked category
STATE CHANGED: <code>selectedCategory = category</code>	DESTINATION: Stays on Screen 2
5. Dish Card Photo / Title Click	
LOCATION: Container 4 (Dish Card)	ACTION: Opens deep item detail and recipe customization screen
STATE CHANGED: <code>selectedDetailItem = item, currentScreen = 3, previousScreen = 2</code>	DESTINATION: Navigates to SCREEN 3 (ITEM CUSTOMIZATION)
6. Quick Customize Sliders Button (Sliders / Gear icon)	
LOCATION: Container 4 (Dish Card)	ACTION: Opens quick customization slide-up bottom drawer without leaving the menu
STATE CHANGED: <code>drawerOpen = true, drawerItem = item</code>	DESTINATION: Opens ItemDrawer modal on Screen 2

7. + ADD Button (Item not in cart, Qty = 0)

LOCATION: Container 4 (Dish Card Bottom Right)

STATE CHANGED: Appends item to cart; Cart Item Count +1, Cart Total +Price

ACTION: Adds 1 unit of dish to cart; triggers tactile haptic feedback

DESTINATION: Stays on Screen 2 (transforms into [-] [1] [+]
stepper)

8. Minus Stepper Button (-, Qty > 0)

LOCATION: Container 4 (Dish Card Stepper)

STATE CHANGED: Updates item quantity in cart; Cart Item Count -1, Cart Total -Price

ACTION: Decrements item quantity by 1 (removes from cart if reaching 0)

DESTINATION: Stays on Screen 2

9. Plus Stepper Button (+, Qty > 0)

LOCATION: Container 4 (Dish Card Stepper)

STATE CHANGED: Updates item quantity in cart; Cart Item Count +1, Cart Total +Price

ACTION: Increments item quantity by 1

DESTINATION: Stays on Screen 2

10. Floating Bottom Cart Pill ("View Cart • ₹{totalCartAmount}")

LOCATION: Container 6 (Floating Dock)

STATE CHANGED: currentScreen = 4, previousScreen = 2

ACTION: Advances to order review

DESTINATION: Navigates to **SCREEN 4 (CART REVIEW)**

SCREEN 3: Item Customization & Flavours (Screen3ItemDetail.tsx & ItemDrawer.tsx)

Purpose: Full culinary configuration view for a selected dish. Lets diners customize Donne Dum Toss style, select add-ons, adjust portions, and view chef spices.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays dish category, dish name, and back button.
- **Container 2: Dish Hero Banner Card:** Large dish photograph, Veg/Non-Veg tag, description, and base price.
- **Container 3: Donne Dum Toss Style / Flavours Card:** Radio selection list for recipe tossing style.
- **Container 4: Add-Ons & Extras Card:** Multi-select checkbox options with extra price indicators.
- **Container 5: Portion & Quantity Card:** Quantity counter stepper with minus/plus buttons.
- **Container 6: Sticky Bottom Action Bar (StickyBottomBar):** Displays dynamic computed price and "Add to Cart" CTA.

Clickable Buttons & Interactive Controls:

1. Header Back Button (Arrow Left icon) LOCATION: Container 1 (WireHeader) STATE CHANGED: currentScreen = 2	ACTION: Returns to menu without committing unadded changes DESTINATION: Navigates back to SCREEN 2 (MENU)
2. Flavour Radio Option Buttons OPTIONS: Authentic Donne Dum, Nati Style Masala Toss, Pepper Garlic Roast STATE CHANGED: selectedOption = choice	ACTION: Selects recipe style with round dot indicator DESTINATION: Stays on Screen 3
3. Add-On Checkbox Buttons OPTIONS: Extra Boiled Egg (+₹20), Biryani Kushka Rice (+₹90), Special Raita & Salna (+₹30) MATH: Dynamically updates currentTotal on CTA button (+ ₹20, +₹90, +₹30)	ACTION: Toggles add-on in selectedAddOns DESTINATION: Stays on Screen 3
4. Quantity Minus Button (-) LOCATION: Container 5 (Quantity Card) MATH: quantity = Math.max(1, quantity - 1)	ACTION: Decrements quantity (minimum 1) DESTINATION: Stays on Screen 3
5. Quantity Plus Button (+) LOCATION: Container 5 (Quantity Card) MATH: quantity = quantity + 1	ACTION: Increments quantity DESTINATION: Stays on Screen 3
6. Sticky Bottom CTA — "Add to Cart • ₹{currentTotal}" LOCATION: Container 6 (StickyBottomBar) STATE CHANGED: Appends item to cart, currentScreen = 4, previousScreen = 3	ACTION: Commits configured dish with selected flavours and add-ons into master cart DESTINATION: Navigates to SCREEN 4 (CART REVIEW)

SCREEN 4: Cart & Order Review (Screen4Cart .tsx)

Purpose: Order verification and dining dispatch hub. Review item list, modify portions, dispatch priority starters separately, or fire complete table order to kitchen.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays table and chair coordinates, back button, and call waiter button.
- **Container 2: Dine-In Context Banner:** Shows Table T-01, Chair C-01, and total dishes selected.
- **Container 3: Selected Dishes List:** Individual item cards with names, quantities, line totals, and priority options.
- **Container 4: Add More Dishes Action Container:** Dashed outline button to browse more items.
- **Container 5: Sticky Bottom Order Dispatch Bar (StickyBottomBar):** Houses the master "Send Order to Kitchen" hero CTA button.

Clickable Buttons & Interactive Controls:

1. Header Back Button LOCATION: Container 1 (WireHeader) STATE CHANGED: currentScreen = 2	ACTION: Returns to menu DESTINATION: Navigates to SCREEN 2 (MENU)
2. Item Quantity Stepper (- and +) LOCATION: Container 3 (Dish Card) MATH: Updates cart totals and counts in real time	ACTION: Modifies item quantity directly in cart DESTINATION: Stays on Screen 4
3. Remove Item Button (Trash icon) LOCATION: Container 3 (Dish Card) STATE CHANGED: Filters item out of cart	ACTION: Removes dish completely from cart DESTINATION: Stays on Screen 4
4. "Order Separately" Priority Button LOCATION: Container 3 (Dish Card) STATE CHANGED: Creates separate tracking entry with status Sent to Kitchen Separately	ACTION: Flags starter dish to be fired independently to kitchen ahead of main biryani courses DESTINATION: Stays on Screen 4
5. "Add More Dishes from Menu" Button (Dashed outline) LOCATION: Container 4 STATE CHANGED: currentScreen = 2	ACTION: Returns to menu to pick more items DESTINATION: Navigates to SCREEN 2 (MENU)
6. Sticky Bottom CTA — "Send Order to Kitchen" LOCATION: Container 5 (StickyBottomBar) STATE CHANGED: orderStage = 'PLACED', currentScreen = 5, previousScreen = 4	ACTION: Dispatches KOT tickets to Supabase, emits KDS broadcast, updates Waiter console table to OCCUPIED, marks items isOrdered = true DESTINATION: Navigates to SCREEN 5 (LIVE TRACKING)

SCREEN 5: Live Order Tracking (Screen5LiveTracking.tsx)

Purpose: Live culinary tracking pipeline synchronized directly with Kitchen Display System (KDS) bump bar.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays Table T-01 • C-01 badge and Call Waiter button.
- **Container 2: 4-Stage Overall Progress Stepper Banner:** Progress bar tracking stages: Received (0%) → Preparing (33%) → Ready (66%) → Collected (100%).
- **Container 3: Individual Item Status Cards:** Displays each ordered dish with cooking stage badge (Received, Preparing, Ready, Collected).
- **Container 4: Add More Dishes to Same Bill Container:** Button to add more items during meal without opening a new bill.
- **Container 5: Sticky Bottom Payment CTA Bar (StickyBottomBar):** Houses "Proceed to Payment" button.

Clickable Buttons & Interactive Controls:

1. "Mark as Collected" Button (Appears when dish is Ready)

LOCATION: Container 3 (Dish Card) **ACTION:** Confirms Floor Captain has delivered the dish to table
STATE CHANGED: Sets item stage to COLLECTED (100% progress) **DESTINATION:** Stays on Screen 5

2. "Add More Dishes to Same Bill" Button

LOCATION: Container 4 **ACTION:** Returns to menu to add extra dishes to the same bill
STATE CHANGED: currentScreen = 2 **DESTINATION:** Navigates to **SCREEN 2 (MENU)**

3. Sticky Bottom CTA — "Proceed to Payment"

LOCATION: Container 5 (StickyBottomBar) **ACTION:** Advances diner to bill summary
STATE CHANGED: currentScreen = 6, previousScreen = 5 **DESTINATION:** Navigates to **SCREEN 6 (ORDER SUMMARY & BILL)**

SCREEN 6: Order Summary & Bill Breakdown (Screen6PaymentBreakdown.tsx)

Purpose: Pre-payment tax breakdown, staff gratuity selection, and multi-diner bill splitting.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays back button returning to live tracking, Table T-01 • C-01, and Call Waiter button.
- **Container 2: Itemized Bill Breakdown Card:** List of dishes with quantities and prices.
- **Container 3: Statutory Tax & Charges Card:** Subtotal, 5% GST (2.5% CGST + 2.5% SGST), and Loyalty Rewards discount.
- **Container 4: Staff Tip Gratuity Card:** Preset tip buttons and custom input.
- **Container 5: Split Bill Calculator Card:** Toggle switch and diner count selector (2 to 5 diners).
- **Container 6: Sticky Bottom Proceed to Payment Bar (StickyBottomBar):** Displays grand total and navigation button.

Clickable Buttons & Interactive Controls:

1. Header Back Button	
LOCATION: Container 1 (WireHeader)	ACTION: Returns to live tracking
STATE CHANGED: currentScreen = 5	DESTINATION: Navigates to SCREEN 5 (LIVE TRACKING)
2. Tip Preset Buttons (₹30, ₹50, ₹100)	
LOCATION: Container 4 (Staff Tip Card)	ACTION: Sets staff tip amount and recalculates grand total
STATE CHANGED: payment.tipAmount = N	MATH: Grand Total = Subtotal + Tax + N - Discount
3. Custom Tip Input Field	
LOCATION: Container 4 (Staff Tip Card)	ACTION: Allows diner to type any custom tip amount
STATE CHANGED: Updates payment.tipAmount	DESTINATION: Stays on Screen 6
4. "Enable Split" Toggle Button	
LOCATION: Container 5 (Split Bill Card)	ACTION: Toggles multi-diner bill calculator
STATE CHANGED: isSplitEnabled = true false	DESTINATION: Stays on Screen 6
5. Split Diners Selector Buttons (2, 3, 4, 5 Diners)	
LOCATION: Container 5 (Split Bill Card)	ACTION: Sets number of diners sharing bill
STATE CHANGED: splitPersons = N	MATH: Recalculates perPersonAmount = Math.ceil(Grand Total / N)
6. Sticky Bottom CTA — "Proceed to Payment (₹{grandTotal})"	
LOCATION: Container 6 (StickyBottomBar)	ACTION: Commits bill breakdown and advances to payment gateway
STATE CHANGED: currentScreen = 7, previousScreen = 6	DESTINATION: Navigates to SCREEN 7 (PAYMENT OPTIONS)

SCREEN 7: Payment Options & Gateway (Screen7PaymentGateway.tsx)

Purpose: Multi-channel payment hub. Supports UPI (Instant dynamic QR & 1-tap app links), Cash (Smart tender notes with return balance calculation), and Card POS terminal request.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays back button returning to bill, Table tag, and Call Waiter button.
- **Container 2: Bill Amount Hero Banner:** Displays grand total payable (₹1,239).
- **Container 3: Payment Method Tabs Row:** Three single-icon, single-word tabs: UPI, Cash, Card.
- **Container 4: UPI Mode Content:** Dynamic NPCI QR Code card + 1-Tap UPI app launch grid.
- **Container 5: Cash Mode Content (Smart Cash Tender):** Dynamic currency note pills (Exact (₹1239), ₹1250, ₹1300, ₹2000), Tender Collection summary card displaying exact return balance, and Floor Captain arrival notice highlighting the return amount.
- **Container 6: Card Mode Content:** Wireless POS machine instructions card.
- **Container 7: Sticky Bottom Action Bar (StickyBottomBar):** Houses the dynamic single-action payment confirmation button.

Clickable Buttons & Interactive Controls:

1. Header Back Button	
LOCATION: Container 1 (WireHeader)	ACTION: Returns to bill breakdown
STATE CHANGED: currentScreen = 6	DESTINATION: Navigates to SCREEN 6 (BILL BREAKDOWN)
2. Payment Tabs (UPI, Cash, Card)	
LOCATION: Container 3 (Payment Method Tabs)	ACTION: Switches active payment mode
STATE CHANGED: activeTab = 'UPI' 'CASH' 'CARD'	DESTINATION: Stays on Screen 7
3. UPI 1-Tap App Buttons (Google Pay, PhonePe, Paytm, Other UPI)	
LOCATION: Container 4 (UPI Mode)	ACTION: On mobile, directly launches the native UPI app with pre-filled amount via deep links (gpay://upi/pay?..)
STATE CHANGED: Sets selectedApp	DESTINATION: Opens native UPI application
4. Smart Cash Tender Note Pills (Exact (₹1239), ₹1250, ₹1300, ₹2000)	
LOCATION: Container 5 (Cash Mode)	ACTION: Selects currency note for cash settlement
MATH: Calculates return balance: Return = Amount - GrandTotal (e.g. 1250 - 1239 = ₹11)	DISPLAY UPDATE: Highlights return amount in Tender Collection card and Floor Captain arrival alert (₹11 RETURN)
5. Sticky Bottom CTA — "Confirm Payment"	
DYNAMIC LABEL: UPI: Pay ₹1239 via UPI Cash (Exact): Confirm Cash (Exact ₹1239) Cash (Change): Confirm Cash ₹1250 (Return ₹11) Card: Request Card Machine (₹1239)	ACTION: Sets isProcessing = true; calls waiterRecordsPayment(); advances after 700ms verification
STATE CHANGED: payment.paymentMethod = activeTab, currentScreen = 8	DESTINATION: Navigates to SCREEN 8 (CONFIRMED & DIGITAL TAX BILL)

SCREEN 8: Confirmed & Digital Tax Invoice (Screen8Confirmation.tsx)

Purpose: Official post-settlement confirmation. Features 12-digit bank UTR proof, Floor Captain console acknowledgement (Captain Suresh), 3-row structured tax receipt, PDF/WhatsApp dispatch, interactive star rating, and session reset.

Containers & Elements:

- Container 1: Top Header (WireHeader):** Displays Payment Confirmed + Table Tag (T-01 • C-01).
- Container 2: Hero Success Card:** Green Checkmark icon, large hero amount Amount Paid: ₹1,239, 12-digit Bank UTR box (BANK UTR: 4281...), and Floor Captain console acknowledgement badge.
- Container 3: Official Digital Tax Invoice (Receipt):** Header with GSTIN, FSSAI, Tax Invoice Number; 3-row balanced metadata grid (Guaranteed non-wrapping timestamps); itemized dishes list with quantities and line totals; statutory 5% GST breakdown and total paid; Floor Captain hospitality sign-off note.
- Container 4: Dispatch Action Buttons Grid:** Download PDF Bill and Share on WhatsApp buttons.
- Container 5: Interactive Experience Rating Card:** 5 interactive star buttons + centered feedback acknowledgement pill below.
- Container 6: Sticky Bottom Reset Bar (StickyBottomBar):** Houses "Start New Order / Dine Again" CTA.

Clickable Buttons & Interactive Controls:

1. "Download PDF Bill" Button	
LOCATION: Container 4 (Dispatch Grid)	ACTION: Generates offline receipt; button text updates to Downloaded!
STATE CHANGED: downloadMsg = true (auto-resets after 2.2s)	DESTINATION: Stays on Screen 8

2. "Share on WhatsApp" Button	
LOCATION: Container 4 (Dispatch Grid)	ACTION: Sends digital receipt link via WhatsApp; button text updates to Sent on WhatsApp!
STATE CHANGED: shareMsg = true (auto-resets after 2.2s)	DESTINATION: Stays on Screen 8

3. Interactive Star Rating Buttons (Stars 1 to 5)	
DEFAULT STATE: All 5 stars start EMPTY (fill-none text-slate-300); subtitle shows Tap stars to rate	ACTION ON CLICK: Fills stars up to chosen rating with gold amber (fill-amber-400); updates subtitle to {rating} / 5 Stars; slides in centered confirmation badge below: [(✓) Thank you for your feedback!]
STATE CHANGED: rating = star, feedbackSubmitted = true	DESTINATION: Stays on Screen 8

4. Sticky Bottom CTA — "Start New Order / Dine Again"	
LOCATION: Container 6 (StickyBottomBar)	ACTION: Calls resetSession(), clears cart, resets table coordinates
STATE CHANGED: currentScreen = 1	DESTINATION: Returns to SCREEN 1 (WELCOME)

SCREEN 9: Call Waiter Assistance Hub (Screen10WaiterCall.tsx)

Purpose: Rapid assistance summon console accessible from the bell icon on every screen.

Containers & Elements:

- **Container 1: Top Header (WireHeader):** Displays Call Waiter + Table Tag (T-01 • C-01) and Back button.
- **Container 2: Active Ping Status Banner:** Animated pulsing notification showing captain summoned status.
- **Container 3: Rapid Assistance Service Grid:** 4 large service buttons with colorful icons.
- **Container 4: Custom Message Input & Send Container:** Free-text input field and send button.

Clickable Buttons & Interactive Controls:

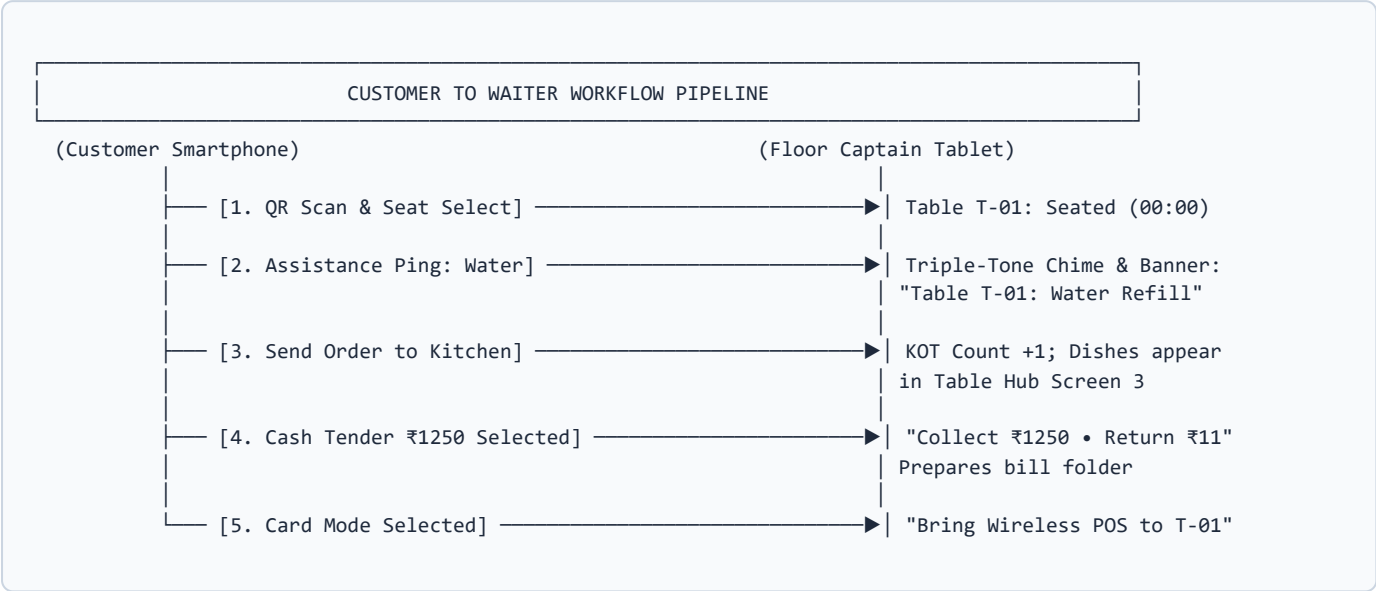
1. Header Back Button (Arrow Left icon) LOCATION: Container 1 (WireHeader) STATE CHANGED: currentScreen = previousScreen	ACTION: Returns diner to their previous active screen (previousScreen 2) DESTINATION: Navigates back to previous screen
2. "Drinking Water" Button (Droplets icon) LOCATION: Container 3 (Service Grid) STATE CHANGED: Adds active water ping	ACTION: Calls pingWaiter('WATER'); emits chime and orange banner on waiter devices DESTINATION: Stays on Screen 9
3. "Extra Tissues" Button (Scroll icon) LOCATION: Container 3 (Service Grid) STATE CHANGED: Adds active tissue ping	ACTION: Calls pingWaiter('TISSUE') DESTINATION: Stays on Screen 9
4. "Cutlery & Salad" Button (Utensils icon) LOCATION: Container 3 (Service Grid) STATE CHANGED: Adds active cutlery ping	ACTION: Calls pingWaiter('CUTLERY') DESTINATION: Stays on Screen 9
5. "Table Cleaning" Button (Sparkles icon) LOCATION: Container 3 (Service Grid) STATE CHANGED: Adds active cleaning ping	ACTION: Calls pingWaiter('TABLE CLEAN') DESTINATION: Stays on Screen 9
6. Custom Message Send Button (Send icon) LOCATION: Container 4 (Custom Message) STATE CHANGED: Sends text payload to Floor Captain	ACTION: Dispatches custom text request via pingWaiter('GENERAL CALL', text) DESTINATION: Stays on Screen 9

PERSISTENT GLOBAL COMPONENTS

- **WireHeader (WireHeader.tsx):** Fixed top app bar present on Screens 2 through 8. Renders Table Tag (T-01 • C-01), Screen Title, Back Button (when applicable), Call Waiter bell trigger (Screen 9), and Cart button with badge counter.
- **StickyBottomBar (StickyBottomBar.tsx):** Fixed bottom CTA dock. Provides single-action flow lock with tactile spring animations, safe-area inset padding for edge-to-edge mobile screens, and contextual total price displays.

SECTION 2: Customer to Waiter Connections

Documents every touchpoint where a diner action triggers real-time sound chimes, banner alerts, status mutations, or order additions on the Floor Waiter's tablet or mobile phone.

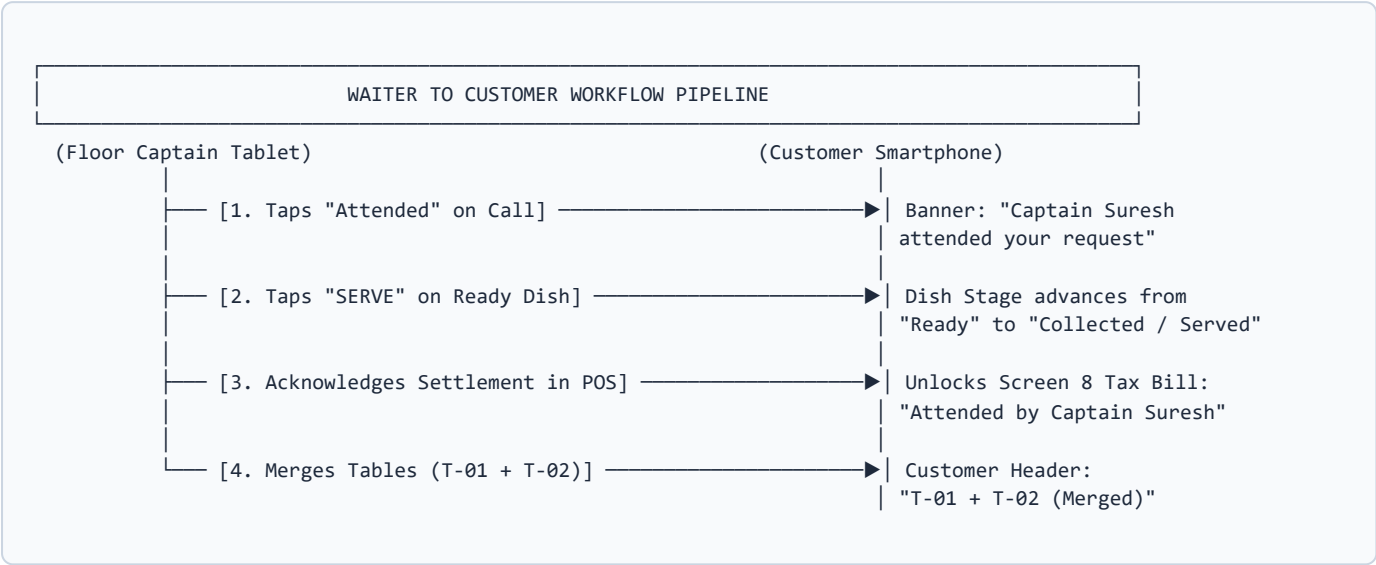


Detailed Customer-to-Waiter Button Actions:

- 1. QR Scan & Chair Select (Screen 1):**
Sets Table T-01 to active. The Floor Captain tablet displays table occupancy and starts the seated dining timer from 00:00.
- 2. Four Assistance Pings & Custom Text (Screen 9):**
Emits a real-time event via BroadcastChannel('thoogudeepa_bridge_sync'). The waiter phone buzzes with a distinct triple-tone sound. A high-priority orange banner drops down on the waiter screen: "Guest Call • Table T-01: Drinking Water Refill".
- 3. Send Order to Kitchen (Screen 4):**
Increments running bill on waiter's console. Dishes populate in waiter's **Table Hub (Screen 3)** with Received or Preparing badges.
- 4. Cash Tender Selection (Screen 7):**
Notifies Floor Captain console of the exact cash note selected (₹1,250) and the exact balance to return (₹11). The Floor Captain prepares the printed bill folder and cash change before walking to the table.
- 5. Request Card Machine (Screen 7):**
Sends notification to Floor Captain to bring the portable wireless POS terminal to Table T-01.

SECTION 3: Waiter to Customer Connections

Documents how Floor Waiter terminal operations communicate back to the diner's mobile screen in real time.

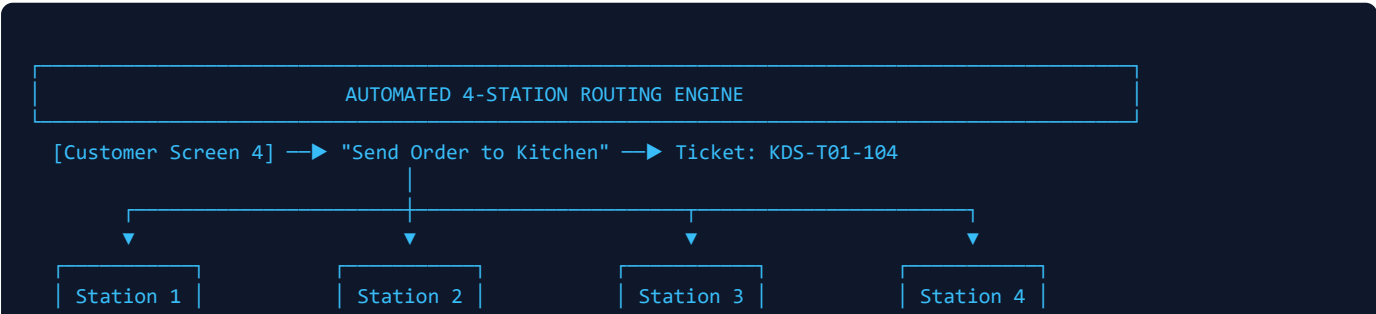


Detailed Waiter-to-Customer Updates:

- Waiter Taps "Attended" on Assistance Ping:**
The pending call is cleared on the waiter terminal. The customer's Screen 9 status banner updates to: "Captain Suresh attended your request. Thank you!".
- Waiter Taps "SERVE" on Ready Dishes:**
When dishes are placed on the dining table, the captain taps SERVE. On the customer's Screen 5, the dish progress bar reaches 100%, and the status badge transitions from Ready to Collected / Served.
- Waiter Records Payment in POS:**
The captain confirms receipt of cash or card authorization. Customer Screen 8 unlocks the official Digital Tax Invoice with the verified badge: Attended by Captain Suresh • Console Acknowledged • Seat Released.
- Waiter Merges Tables (T-01 + T-02):**
If two parties decide to sit together, the captain merges tables on the tablet. On the customer's smartphone, the header tag updates to T-01 + T-02 (Merged), and the combined dishes reflect on the shared bill.

SECTION 4: Customer to Kitchen Connections

Documents how orders placed on the diner's mobile phone are automatically sorted, partitioned, and routed to four dedicated kitchen stations.



Dum Biryani	Tandoor & Starters	Curry & Breads	Beverages & Dessert
• Donne Mutton Biryani • Chicken Dum Biryani	• Chicken Ghee Roast • Mutton Nalli Fry	• Mutton Gravy Curry • Roti / Parotta	• Elaneer Payasam • Fresh Lime Soda

Detailed Customer-to-Kitchen Process:

1. **Firing Order via `placeAllOrders()` (Screen 4):** Packages item names, quantities, portion choices, and recipe preferences. Generates sequential ticket ID: KDS-T01-104. Dispatches payload instantly across `BroadcastChannel('thoogudeepa_bridge_sync')`.
2. **Priority Starter Firing ("Order Separately"):** If a customer taps "Order Separately" on a starter (e.g. Pepper Chicken), the ticket is routed with a high-priority flag so the starter chef prepares it immediately while main biryanis simmer.
3. **Culinary Modifiers & Chef Notes:** Donne Dum Toss Style (Nati Style Masala Toss) and Add-ons (Extra Boiled Egg) print directly onto the station KDS screen so chefs prepare the exact customized recipe.

SECTION 5: Kitchen to Customer Updates

Documents the 4-stage cooking pipeline progression and out-of-stock (86) inventory lockouts.

5.1 The 4-Stage Bump Bar Progression:

4-STAGE KITCHEN BUMP BAR PROGRESSION		
Stage 1: REC (Received)	→ 0% Progress	(Gray clock icon on Customer Screen 5)
Stage 2: PREP (Preparing)	→ 33% Progress	(Amber flame icon on Customer Screen 5)
Stage 3: READY (Plated)	→ 66% Progress	(Green pulsing badge; "Mark as Collected" unlocks)
Stage 4: SERVED (Collected)	→ 100% Progress	(Checkmark badge; Dish completed)

- Stage 1 — Received (REC):** The ticket lands on the chef's KDS screen. On the customer's phone (Screen 5), the item displays a gray clock icon with status Received (0% progress).
- Stage 2 — Preparing (PREP):** The chef begins cooking and taps PREP. The customer's item card turns amber with a flame icon and status Preparing (33% progress).
- Stage 3 — Ready / Plated (READY):** The chef garnishes the dish and taps the green READY button on the KDS. The customer's Screen 5 card pulses with a vibrant emerald badge Ready (66% progress), and the tactile "Mark as Collected" button unlocks.
- Stage 4 — Served / Collected (COLLECTED):** Either the Floor Captain taps SERVE or the customer taps Mark as Collected. The item status updates to Collected / Served (100% progress).

5.2 Kitchen 86 (Sold-Out / Out of Stock) Instant Lockout:

When the head chef runs out of an ingredient (e.g. *Mutton Nalli Fry* at 02:15 PM):

- Chef Taps 86 Toggle:** On the kitchen tablet (/kitchen), the chef taps the 86 / Sold Out toggle next to that dish.
- Instant Broadcast:** The event is broadcast across BroadcastChannel('thoogudeepa_bridge_sync') with payload { id: 'item-3', is86: true }.
- Immediate Lockout on Customer Phone:**
 - On Screen 2 (Menu):** The dish card instantly dims into a disabled state with a bold red badge 86 - Sold Out. The + ADD button is disabled and locked to prevent touches.
 - On Screen 3 (Item Detail):** The "Add to Cart" button is disabled and grayed out with a Ban icon.
- Restoration:** As soon as a fresh batch is prepared, the chef untoggles 86 on the KDS, and the dish instantly unlocks across all customer devices.

SECTION 6: Master Button-by-Button Comprehensive Reference Matrix

Exhaustive dictionary of every single clickable control, button, stepper, and toggle across all 9 screens in the customer portal.

Button Name / Label	Screen & Location	Action When Pressed	State Variable Changed	Mathematical / Counter Updates	Next Navigation Destination
Chair Button (C-01 to C-04)	Screen 1 (Grid)	Selects specific dining seat at scanned table	seatNumber = N	N/A	Stays on Screen 1
Connect 5G Wi-Fi	Screen 1 (Hero CTA)	Authenticates session via local restaurant Wi-Fi	currentScreen = 2	N/A	Navigates to Screen 2
Continue Mobile Data	Screen 1 (Secondary)	Enters menu using cellular connection	currentScreen = 2	N/A	Navigates to Screen 2
Search Clear (X icon)	Screen 2 (Search Bar)	Clears active dish search input text	searchTerm = ""	Restores full dish list	Stays on Screen 2
Category Filter Pill	Screen 2 (Tabs)	Filters menu dishes by culinary category	selectedCategory = cat	Updates visible cards	Stays on Screen 2
Dish Card Photo / Title	Screen 2 (Menu Grid)	Opens full detail page for that dish	selectedDetailItem = item	N/A	Navigates to Screen 3
Sliders Customizer (Opt)	Screen 2 (Dish Card)	Opens quick customization slide-up sheet	drawerOpen = true	N/A	Opens ItemDrawer on Screen 2
+ ADD Button (Qty = 0)	Screen 2 (Dish Card)	Adds 1 quantity of dish to cart	Appends item to cart	Cart count +1, Total +Price	Stays on Screen 2
Minus Stepper (Qty > 0)	Screen 2 (Dish Card)	Decreases dish quantity in cart	Modifies item in cart	Cart count -1, Total -Price	Stays on Screen 2
Plus Stepper (Qty > 0)	Screen 2 (Dish Card)	Increases dish quantity in cart	Modifies item in cart	Cart count +1, Total +Price	Stays on Screen 2
Floating View Cart Bar	Screen 2 (Bottom Dock)	Opens order cart review	currentScreen = 4	N/A	Navigates to Screen 4
Back Button (Arrow Left)	Screen 3 (Header)	Returns to menu without adding dish	currentScreen = 2	N/A	Navigates to Screen 2
Flavour Radio Option	Screen 3 (Body)	Selects Donne Dum style / flavour	selectedOption = choice	N/A	Stays on Screen 3
Add-On Checkbox	Screen 3 (Body)	Toggles optional egg/kushka/salna	Toggles in selectedAddOns	Updates currentTotal	Stays on Screen 3
Add to Cart CTA	Screen 3 (Bottom Bar)	Commits customized dish into cart	Appends item to cart	Cart count +Qty, Total +Subtotal	Navigates to Screen 4
Order Separately	Screen 4 (Cart Card)	Flags starter to be prepared ahead of meal	Adds priority tracking record	Priority KDS flag	Stays on Screen 4
Remove Dish (Trash icon)	Screen 4 (Cart Card)	Removes dish completely from cart	Removes item from cart	Cart count -N, Total -Price	Stays on Screen 4
Add More Dishes Button	Screen 4 & 5 (Body)	Returns to menu to pick more items	currentScreen = 2	N/A	Navigates to Screen 2
Send Order to Kitchen	Screen 4 (Bottom Bar)	Fires KOT tickets to Supabase & KDS	isOrdered = true, orderStage = 'PLACED'	Sets table to OCCUPIED	Navigates to Screen 5
Mark as Collected	Screen 5 (Item Card)	Confirms ready dish delivered to table	Item stage = COLLECTED	Ready items count -1	Stays on Screen 5

Button Name / Label	Screen & Location	Action When Pressed	State Variable Changed	Mathematical / Counter Updates	Next Navigation Destination
Proceed to Payment	Screen 5 (Bottom Bar)	Opens bill breakdown & tip selection	currentScreen = 6	N/A	Navigates to Screen 6
Tip Preset (₹30, ₹50, ₹100)	Screen 6 (Body)	Adds staff gratuity to grand total	payment.tipAmount = N	Recalculates Grand Total	Stays on Screen 6
Enable Split Toggle	Screen 6 (Body)	Toggles multi-diner bill calculation	isSplitEnabled = bool	Calculates ₹ {perPerson}	Stays on Screen 6
Split Diners (2 to 5)	Screen 6 (Body)	Sets number of diners sharing bill	splitPersons = N	Recalculates ₹ {perPerson}	Stays on Screen 6
Proceed to Payment CTA	Screen 6 (Bottom Bar)	Advances to payment methods selection	currentScreen = 7	N/A	Navigates to Screen 7
Payment Tab (UPI)	Screen 7 (Tabs)	Activates dynamic QR & deep links mode	activeTab = 'UPI'	N/A	Stays on Screen 7
Payment Tab (Cash)	Screen 7 (Tabs)	Activates physical cash tender mode	activeTab = 'CASH'	N/A	Stays on Screen 7
Payment Tab (Card)	Screen 7 (Tabs)	Activates table card machine request	activeTab = 'CARD'	N/A	Stays on Screen 7
1-Tap App (GPay/PhonePe)	Screen 7 (UPI Grid)	Launches mobile UPI payment intent	selectedApp = name	Pre-fills amount in app	Opens UPI app
Tender Pill (Exact/Notes)	Screen 7 (Cash Grid)	Selects currency note for cash payment	selectedTender = opt	Computes exact return balance	Stays on Screen 7
Confirm Payment CTA	Screen 7 (Bottom Bar)	Settles bill & notifies waiter console	currentScreen = 8	Table status = BILLING	Navigates to Screen 8
Download PDF Bill	Screen 8 (Action Grid)	Generates printable digital tax invoice	downloadMsg = true	N/A	Stays on Screen 8
Share on WhatsApp	Screen 8 (Action Grid)	Dispatches digital invoice to WhatsApp	shareMsg = true	N/A	Stays on Screen 8
Star Rating (1 to 5)	Screen 8 (Rating Card)	Rates dining experience (starts empty)	rating = star, feedbackSubmitted = true	Fills stars, shows thank-you pill	Stays on Screen 8
Dine Again / New Order	Screen 8 (Bottom Bar)	Clears session and returns to start	Calls resetSession(), currentScreen = 1	Clears cart & reset totals	Navigates to Screen 1
Call Waiter Bell (Header)	Global (WireHeader)	Opens immediate waiter assistance hub	navigateTo(9)	N/A	Navigates to Screen 9
Quick Call (Water/Tissue)	Screen 9 (Grid)	Sends specific service ping to waiter	Calls pingWaiter(type)	Waiter pending pings +1	Stays on Screen 9
Send Custom Message	Screen 9 (Bottom)	Sends custom text request to captain	Calls pingWaiter('GENERAL', text)	Waiter alert banner created	Stays on Screen 9
Back from Waiter Hub	Screen 9 (Header)	Returns to previous screen	navigateTo(previousScreen)	N/A	Returns to previous active screen